

DEPLOYMENT GUIDE

H2O Online Deployment Guide

Local development with MongoDB Atlas, GitHub workflow, AWS EC2 deployment, production API configuration, and pre-deploy checklist.

13.62.57.255

AWS EC2 IP

:5000

Production API

:3000

Admin portal

13

Sections

Document version: 1.0

API: <http://13.62.57.255:5000>

Admin: <http://13.62.57.255:3000>

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H2O Online — Deployment Guide

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Version: 1.0

Platforms: Local development · GitHub · AWS EC2

Production API (current): `http://13.62.57.255:5000`

Production admin (current): `http://13.62.57.255:3000`

1. Overview

H2O Online has three deployable components:

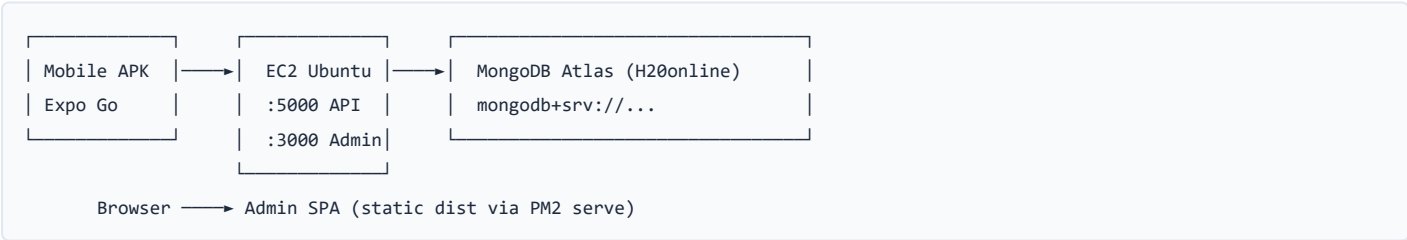
Component	Folder	Local URL	Production (AWS) URL
Backend API	<code>backend/</code>	<code>http://localhost:5000</code>	<code>http://13.62.57.255:5000</code>
Admin portal	<code>admin/</code>	<code>http://localhost:5174</code>	<code>http://13.62.57.255:3000</code>
Mobile app	<code>mobile/</code>	Expo Metro :8081	APK → production API

Database: MongoDB Atlas (`H2Oonline` database) — used for both local dev and production.

Single source of truth for AWS URLs: `config/aws-production.json`

When the EC2 IP or domain changes, update that file **and** sync all files listed in Section 3.

2. Architecture (Production)



GitHub flow: Push code → SSH to EC2 → `git pull` → rebuild → `pm2 restart`

3. Configuration matrix — what to change where

3.1 [△](#) CHANGE FOR PRODUCTION (before AWS deploy)

#	File	Variable / setting	Production value	Notes
1	<code>backend/.env</code> (on server)	<code>NODE_ENV</code>	<code>production</code>	Required — enables CORS lock, env validation
2	<code>backend/.env</code>	<code>MONGODB_URI</code>	Atlas connection string	Same cluster OK; use strong credentials
3	<code>backend/.env</code>	<code>JWT_SECRET</code>	Random 32+ bytes	<code>openssl rand -hex 32</code>
4	<code>backend/.env</code>	<code>MASTER_ADMIN_PASSWORD</code>	Strong password	Must not be <code>admin@H2O</code>
5	<code>backend/.env</code>	<code>ALLOWED_ORIGINS</code>	<code>http://13.62.57.255:3000</code>	Admin browser origin; add HTTPS domain later
6	<code>backend/.env</code>	<code>RAZORPAY_KEY_ID</code>	Live key	<code>rzp_live_...</code> for real payments
7	<code>backend/.env</code>	<code>RAZORPAY_KEY_SECRET</code>	Live secret	Never commit to GitHub

#	File	Variable / setting	Production value	Notes
8	admin/.env.production	VITE_API_URL	http://13.62.57.255:5000	Baked into admin build
9	mobile/config/apiUrl.json	production	http://13.62.57.255:5000	APK fallback URL
10	mobile/eas.json	EXPO_PUBLIC_API_URL	http://13.62.57.255:5000	All build profiles
11	config/aws-production.json	all fields	Current EC2 IP/URLs	Central reference
12	AWS Security Group	Inbound rules	22, 5000, 3000, 80, 443	See Section 7

Template files (safe to commit):

- Local backend: backend/.env.local.example → copy to backend/.env
- Production backend: backend/.env.production.example → copy to backend/.env on server
- Local admin: admin/.env.example → copy to admin/.env
- Production admin: admin/.env.production.aws.example → copy to admin/.env.production
- Local mobile: mobile/.env.local.example → copy to mobile/.env

3.2 Local development (no production flags)

File	Setting	Local value
backend/.env	NODE_ENV	development
backend/.env	MONGODB_URI	Atlas URI (same as prod cluster OK)
backend/.env	PORT	5000
admin/.env	VITE_API_URL	http://localhost:5000
mobile/.env	EXPO_PUBLIC_API_URL	http://YOUR_PC_IP:5000 or run npm run local

Do not set NODE_ENV=production locally unless testing CORS/production validation.

4. Run locally (Atlas DB + local servers)

Prerequisites

- Node.js 18+
- Git
- MongoDB Atlas cluster reachable from your PC (IP allowlist: 0.0.0.0/0 for dev or your IP)
- Phone on same Wi-Fi for Expo Go (optional)

Step 1 — Backend

```
cd backend
npm install
cp .env.local.example .env
# Edit .env: set MONGODB_URI (Atlas), JWT_SECRET, Razorpay test keys, Google Maps key
npm run check-mongo
npm run dev
```

Verify: http://localhost:5000/api/health → { "ok": true, "db": "connected", "env": "development" }

Step 2 — Admin

```
cd admin
npm install
cp .env.example .env
npm run dev
```

Open: `http://localhost:5174`

Login: `H2O admin / admin@H2O`

Step 3 — Mobile

```
cd mobile
npm install
npm run local
```

This auto-sets `EXPO_PUBLIC_API_URL` to your PC LAN IP and starts Expo. Scan QR with Expo Go.

Android emulator: set `EXPO_PUBLIC_API_URL=http://10.0.2.2:5000` in `mobile/.env`

Test production API from Expo (no APK rebuild):

```
cd mobile
npm run prod:local
npx expo start -c
```

5. GitHub setup

5.1 Repository

1. Create a GitHub repository (private recommended).
2. Ensure `.gitignore` excludes secrets:

- - `backend/.env`
- - `admin/.env` , `admin/.env.production`
- - `mobile/.env`

1. Push the codebase:

```
git add .
git commit -m "Initial H2O Online platform"
git remote add origin https://github.com/YOUR_ORG/H2Oonline.git
git push -u origin main
```

Never commit real `MONGODB_URI` , `JWT_SECRET` , Razorpay live keys, or `.env` files.

5.2 Branch strategy (recommended)

Branch	Purpose
<code>main</code>	Production-ready code deployed to EC2
<code>develop</code>	Integration / testing (optional)

Deploy to AWS from `main` after testing locally.

5.3 What stays on the server only (not in GitHub)

Secret / config	Location
Production <code>backend/.env</code>	EC2 <code>/home/ubuntu/H2Oonline/backend/.env</code>
Built admin <code>dist/</code>	Generated on server via <code>npm run build:prod</code>
PM2 process list	<code>pm2 save</code> on EC2

6. AWS EC2 deployment

6.1 Server requirements

- Ubuntu 22.04 LTS (or similar)
- Node.js 18+ (`nvm install 18`)
- PM2: `sudo npm install -g pm2 serve`
- Git
- Ports open in security group (Section 7)

6.2 First-time setup on EC2

```
# SSH into EC2
ssh -i your-key.pem ubuntu@13.62.57.255

# Clone repository
cd /home/ubuntu
git clone https://github.com/YOUR_ORG/H2Oonline.git
cd H2Oonline

# Backend
cd backend
npm install --omit=dev
cp .env.production.example .env
nano .env # Set MONGODB_URI, JWT_SECRET, MASTER_ADMIN_PASSWORD, ALLOWED_ORIGINS, Razorpay keys

# Verify MongoDB + env
npm run check-mongo
node scripts/check-razorpay-env.js

# Admin build
cd ../admin
npm install
cp .env.production.aws.example .env.production
npm run build:prod

# Start with PM2 (from repo root)
cd ..
pm2 start ecosystem.config.cjs --env production
pm2 save
pm2 startup # follow printed command
```

6.3 Verify deployment

```
curl http://localhost:5000/api/health
curl http://13.62.57.255:5000/api/health
```

Browser:

- Admin: `http://13.62.57.255:3000`
- API health: `http://13.62.57.255:5000/api/health`

6.4 Redeploy after GitHub push

On EC2:

```

cd /home/ubuntu/H2Oonline
git pull origin main

# Backend
cd backend
npm install --omit=dev
pm2 restart h2o-backend

# Admin (rebuild – API URL is baked in at build time)
cd ../admin
npm install
npm run build:prod
pm2 restart h2o-admin

pm2 save
pm2 status

```

6.5 Mobile APK (production API)

On your dev machine:

```

cd mobile
# Confirm eas.json + config/apiUrl.json point to http://13.62.57.255:5000
npm run build:apk

```

Install APK on devices. API URL is embedded at build time.

7. AWS security group

Port	Protocol	Source	Purpose
22	TCP	Your IP	SSH
5000	TCP	0.0.0.0/0	Backend API (mobile + admin)
3000	TCP	0.0.0.0/0	Admin SPA (PM2 serve)
80	TCP	0.0.0.0/0	Nginx HTTP (optional)
443	TCP	0.0.0.0/0	Nginx HTTPS (optional)

MongoDB Atlas: Network Access → allow EC2 public IP or `0.0.0.0/0` (restrict in production if possible).

8. Production API reference

Base URL: `http://13.62.57.255:5000`

Auth: Authorization: Bearer <JWT> (customer/supplier/delivery routes)

8.1 Health & public

Method	Path	Auth
GET	<code>/api/health</code>	No

8.2 Customer mobile APIs

Prefix	Examples
<code>/api/auth</code>	login, register

Prefix	Examples
/api/users	me, update profile
/api/orders	create, list, track
/api/water-intake	log intake, summary
/api/leaderboard	monthly rankings, preferences
/api/wallet	balance, credit, debit
/api/subscriptions	list, create, cancel
/api/payments	Razorpay create-order, verify
/api/settings	tax, payment settings
/api/serviceability	delivery area check
/api/addresses	CRUD
/api/bills	list, pay
/api/ai	water insight, report, ask

8.3 Admin portal APIs

Base: same host · **Prefix:** /api/admin/*

Auth: Admin JWT from POST /api/admin/auth/login

Area	Paths
Users	/api/admin/users , CRUD
Orders	/api/admin/orders
Suppliers	/api/admin/suppliers , verify
Delivery partners	/api/admin/delivery-partners
Plans & products	/api/admin/plans , plan-products
Subscriptions	/api/admin/subscriptions
Financials	/api/admin/financials
Wallet	/api/admin/wallet-management
Serviceable areas	/api/admin/serviceable-areas
Tax settings	/api/admin/tax-settings
Surveys	/api/admin/surveys

8.4 Mobile client mapping

App	Config file	Production value
APK / release	mobile/eas.json	EXPO_PUBLIC_API_URL=http://13.62.57.255:5000
APK fallback	mobile/config/apiUrl.json	"production": "http://13.62.57.255:5000"
Admin SPA	admin/.env.production	VITE_API_URL=http://13.62.57.255:5000

9. Pre-deployment checklist

Backend

- ☐ NODE_ENV=production on EC2
- ☐ Strong JWT_SECRET (not default)
- ☐ MASTER_ADMIN_PASSWORD changed

- ☐ `ALLOWED_ORIGINS` includes admin URL
- ☐ Atlas URI correct; EC2 IP allowed in Atlas
- ☐ Razorpay keys set (live for production payments)
- ☐ `curl http://13.62.57.255:5000/api/health` returns ok

Admin

- ☐ `admin/.env.production` has production `VITE_API_URL`
- ☐ `npm run build:prod` succeeds
- ☐ PM2 serving `admin/dist` on port 3000
- ☐ Login works at `http://13.62.57.255:3000`

Mobile

- ☐ `eas.json` production profile URL correct
- ☐ `config/apiUrl.json` matches
- ☐ New APK built after API URL change
- ☐ Security group port 5000 open

GitHub

- ☐ No `.env` files committed
- ☐ Latest code pushed to `main`
- ☐ EC2 `git pull` completed

10. Troubleshooting

Issue	Fix
CORS error on admin	Set <code>ALLOWED_ORIGINS=http://13.62.57.255:3000</code> and <code>NODE_ENV=production</code> ; restart backend
Mobile cannot reach API	Check security group port 5000; verify URL in <code>mobile/.env</code>
<code>db: disconnected</code>	Check Atlas URI; add EC2 IP to Atlas Network Access
Admin shows wrong API	Rebuild admin with correct <code>.env.production</code>
Payments fail	Run <code>node backend/scripts/check-razorpay-env.js</code> ; use matching test/live keys on backend + mobile
Production env check failed	See error list; fix <code>JWT_SECRET</code> , <code>MASTER_ADMIN_PASSWORD</code> , <code>ALLOWED_ORIGINS</code>

11. HTTPS & custom domain (optional)

See `docs/DOMAIN-SETUP.md` for Nginx + Let's Encrypt.

After HTTPS:

1. Update `config/aws-production.json`
2. Update `backend/.env` → `ALLOWED_ORIGINS=https://admin.yourdomain.com`
3. Rebuild admin with `VITE_API_URL=https://api.yourdomain.com`
4. Rebuild mobile APK with new URL in `eas.json`
5. Open ports 80/443; consider closing direct 5000/3000 to public

12. Quick command reference

Local

```
cd backend && npm run dev
cd admin && npm run dev
cd mobile && npm run local
```

Production (EC2)

```
cd /home/ubuntu/H2Online
git pull
cd backend && npm install --omit=dev && pm2 restart h2o-backend
cd ../admin && npm run build:prod && pm2 restart h2o-admin
pm2 status
```

Generate this PDF

```
cd docs
npm run deployment-guide
```

13. Document control

Version	Date	Changes
1.0	June 2026	Initial deployment guide — AWS EC2 13.62.57.255, Atlas, GitHub flow

End of document — H2O Online Deployment Guide v1.0